



Participante: Luis Ramón Reyes Volquez

Técnico: Ciberseguridad

Módulo: Análisis Forense

Facilitador: Kevin Feliz Henríquez

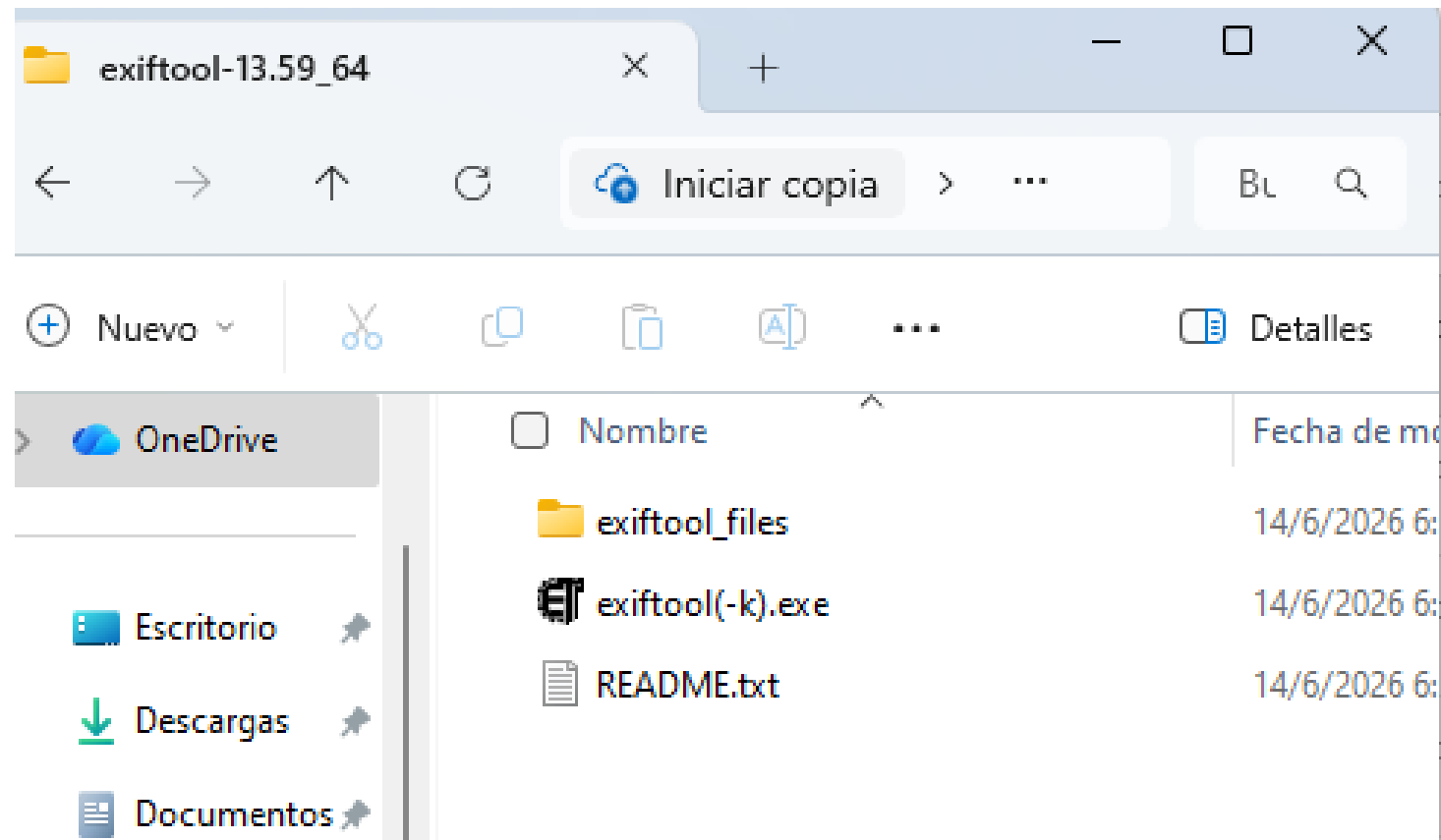
Centro: Cyberclass - INFOTEP

Práctica: Encontrar metadatos de imágenes y documentos con Exiftool.

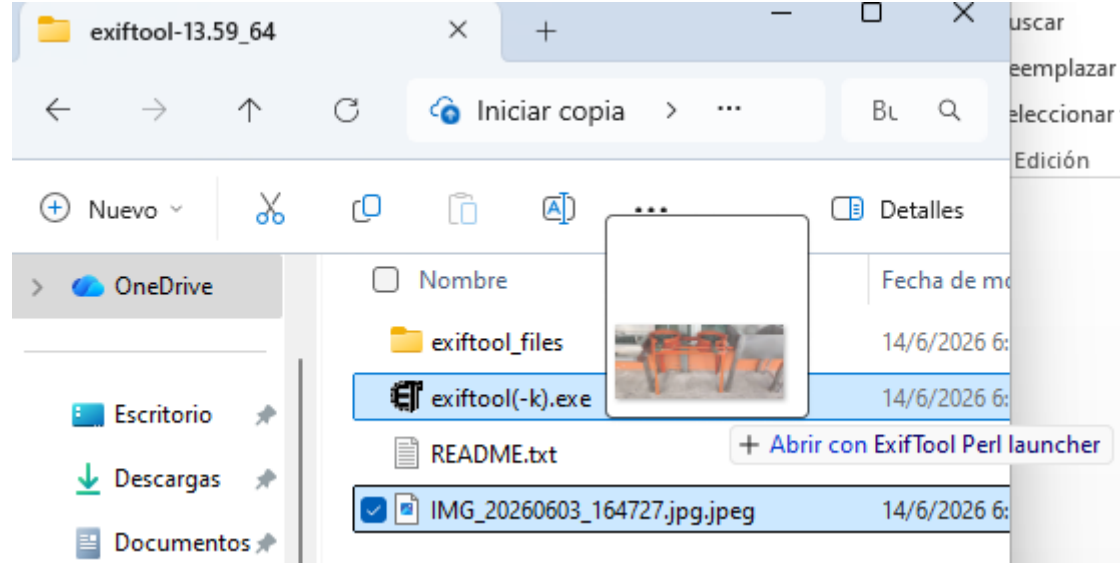
Descripción: Practica de como utilizar el programa Exiftool para la obtención de metadatos.

Link de GitHub: <https://github.com/elchecho3000-blip/Ciberseguridad-bootcamp/tree/main/An%C3%A1lisis%20Forense>

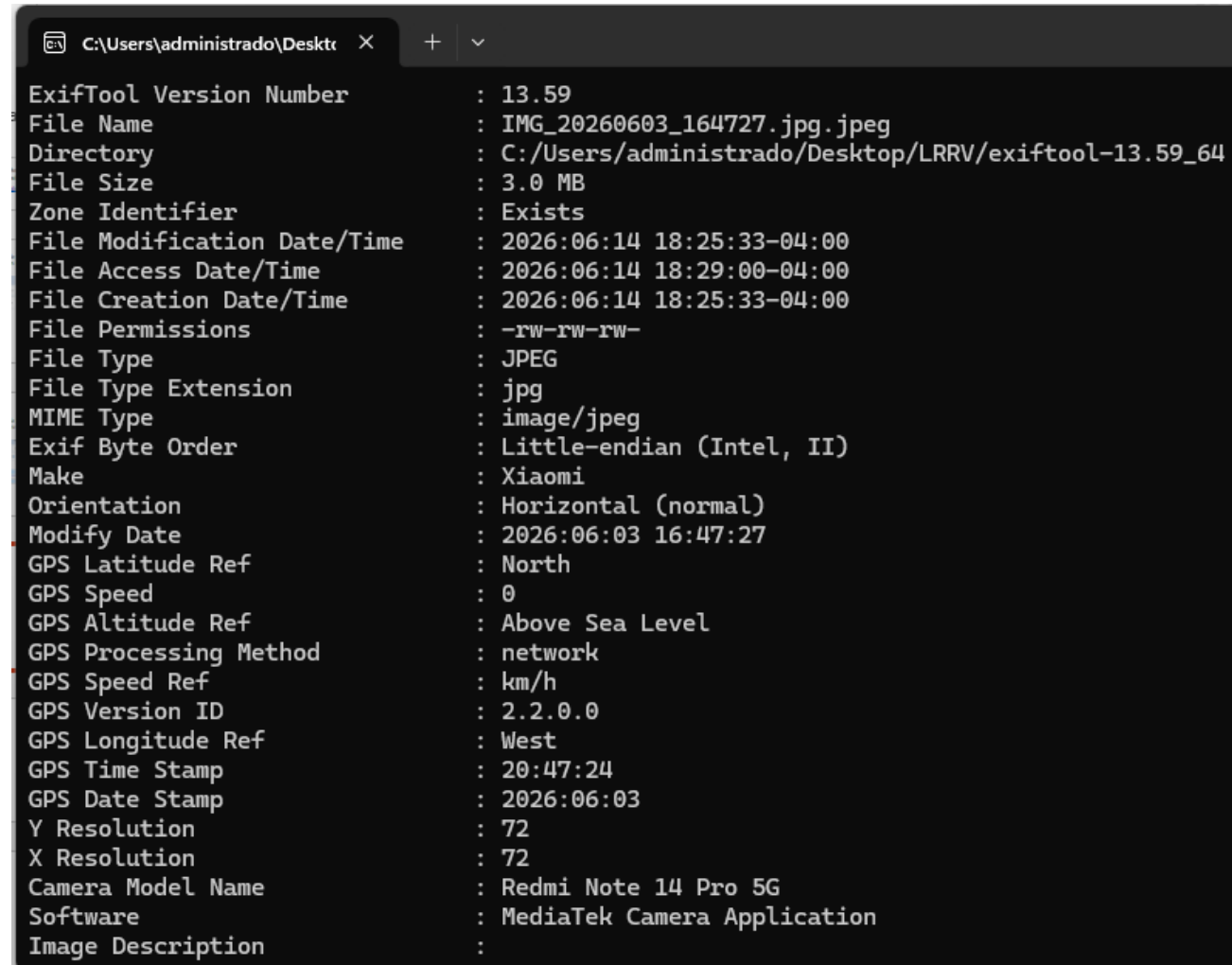
Este es el programa Exiftool el cual puede utilizarse vía descarga o vía web, en esta ocasión lo utilizaremos modo de descarga



Solo tenemos que arrastrar el documento
a la aplicación para el escaneo



Y listo, obtendremos los metadatos del primer archivo



```
C:\Users\administrado\Desktop > exiftool IMG_20260603_164727.jpg.jpeg

ExifTool Version Number      : 13.59
File Name                    : IMG_20260603_164727.jpg.jpeg
Directory                   : C:/Users/administrado/Desktop/LRRV/exiftool-13.59_64
File Size                    : 3.0 MB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:06:14 18:25:33-04:00
File Access Date/Time        : 2026:06:14 18:29:00-04:00
File Creation Date/Time      : 2026:06:14 18:25:33-04:00
File Permissions              : -rw-rw-rw-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Make                         : Xiaomi
Orientation                  : Horizontal (normal)
Modify Date                   : 2026:06:03 16:47:27
GPS Latitude Ref              : North
GPS Speed                     : 0
GPS Altitude Ref              : Above Sea Level
GPS Processing Method         : network
GPS Speed Ref                 : km/h
GPS Version ID                : 2.2.0.0
GPS Longitude Ref             : West
GPS Time Stamp                : 20:47:24
GPS Date Stamp                : 2026:06:03
Y Resolution                  : 72
X Resolution                  : 72
Camera Model Name             : Redmi Note 14 Pro 5G
Software                     : MediaTek Camera Application
Image Description              :
```

```
C:\Users\administrado\Desktop X + v
Software : MediaTek Camera Application
Image Description :
Y Cb Cr Positioning : Co-sited
Exif Version : 0220
Aperture Value : 1.6
Mirror : false
Sensor Type : rear
Hdr : off
Op Mode : 36869
Small Picture : false
AI Scene : 0
Filter Id : 66048
Zoom Multiple : 1
Exposure Compensation : 0
Exposure Program : Program AE
Color Space : Uncalibrated
Max Aperture Value : 1.6
Exif Image Height : 1840
Brightness Value : 4.2
Date/Time Original : 2026:06:03 16:47:27
Flashpix Version : 0100
Maker Note Byte Order : Big-endian (Motorola, MM)
Sub Sec Time Original : 683
White Balance : Auto
Interoperability Index : R98 - DCF basic file (sRGB)
Interoperability Version : 0100
Recommended Exposure Index : 0
Exposure Mode : Auto
Exposure Time : 1/120
Offset Time : -04:00
```

```
C:\Users\administrado\Desktop X + v
Exposure Time : 1/120
Offset Time : -04:00
Flash : Auto, Did not fire
Sub Sec Time : 683
F Number : 1.6
Exif Image Width : 4080
ISO : 250
Components Configuration : Y, Cb, Cr, -
Offset Time Digitized : -04:00
Focal Length In 35mm Format : 23 mm
Sub Sec Time Digitized : 683
Digital Zoom Ratio : 0
Create Date : 2026:06:03 16:47:27
Shutter Speed Value : 1/120
Metering Mode : Center-weighted average
Focal Length : 6.2 mm
Sensitivity Type : ISO Speed
Offset Time Original : -04:00
Scene Capture Type : Standard
Light Source : Other
Resolution Unit : inches
Xiaomi Model : Redmi Note 14 Pro 5G
Compression : JPEG (old-style)
Thumbnail Offset : 1991
Thumbnail Length : 38784
Profile CMM Type :
Profile Version : 4.0.0
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
```

```
C:\Users\administrado\Desktop X + v
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
Profile Date Time : 2022:08:08 08:08:08
Profile File Signature : acsp
Primary Platform : Unknown ()
CMM Flags : Not Embedded, Independent
Device Manufacturer : Xiaomi
Device Model :
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Apple Computer Inc.
Profile ID : 0
Profile Description : sRGB EOTF with DCI-P3 Color Gamut
Profile Copyright : Copyright Xiaomi Communications Co., Ltd. 2022
Media White Point : 0.96419 1 0.82489
Red Matrix Column : 0.51512 0.2412 -0.00105
Green Matrix Column : 0.29198 0.69225 0.04189
Blue Matrix Column : 0.1571 0.06657 0.78407
Chromatic Adaptation : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.01706 -0.00923 0.01508 0.75168
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width : 4080
Image Height : 1840
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
```

```
C:\Users\administrado\Desktop X + v
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width : 4080
Image Height : 1840
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 1.6
Image Size : 4080x1840
Megapixels : 7.5
Scale Factor To 35 mm Equivalent: 3.7
Shutter Speed : 1/120
Create Date : 2026:06:03 16:47:27.683-04:00
Date/Time Original : 2026:06:03 16:47:27.683-04:00
Modify Date : 2026:06:03 16:47:27.683-04:00
Thumbnail Image : (Binary data 38784 bytes, use -b option to extract)
GPS Altitude : 14.4 m Above Sea Level
GPS Date/Time : 2026:06:03 20:47:24Z
GPS Latitude : 18 deg 29' 15.94" N
GPS Longitude : 69 deg 53' 50.38" W
Circle Of Confusion : 0.008 mm
Field Of View : 76.1 deg
Focal Length 35mm Equiv : 6.2 mm (35 mm equivalent: 23.0 mm)
GPS Position : 18 deg 29' 15.94" N, 69 deg 53' 50.38" W
Hyperfocal Distance : 2.87 m
Light Value : 7.0
-- press ENTER --
```

Otra opción de usarse es vía el CMD, analizaremos el segundo archivo

```
GPS Altitude           : 13.7 m Above Sea Level
GPS Date/Time          : 2026:05:15 13:58:33Z
GPS Latitude           : 18 deg 29' 19.73" N
GPS Longitude          : 69 deg 54' 13.14" W
Circle Of Confusion    : 0.008 mm
Field Of View          : 76.1 deg
Focal Length 35mm Equiv : 6.2 mm (35 mm equivalent: 23.0 mm)
GPS Position           : 18 deg 29' 19.73" N, 69 deg 54' 13.14" W
Hyperfocal Distance    : 2.87 m
Light Value            : 3.5

C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>
```



```
CA\ Administrador: Símbolo del sistema

C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool IMG_20260515_095837.jpg
ExifTool Version Number      : 13.59
File Name                    : IMG_20260515_095837.jpg.jpeg
Directory                   : .
File Size                    : 2.8 MB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:06:17 15:15:04-04:00
File Access Date/Time        : 2026:06:17 15:37:47-04:00
File Creation Date/Time      : 2026:06:17 15:15:04-04:00
File Permissions             : -rw-rw-rw-
File Type                   : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order              : Little-endian (Intel, II)
Make                        : Xiaomi
Orientation                  : Unknown (0)
Modify Date                  : 2026:05:15 09:58:38
GPS Latitude Ref             : North
GPS Speed                    : 0
GPS Altitude Ref             : Above Sea Level
GPS Processing Method        : network
GPS Speed Ref                : km/h
GPS Version ID               : 2.2.0.0
GPS Longitude Ref            : West
GPS Time Stamp               : 13:58:33
GPS Date Stamp               : 2026:05:15
Y Resolution                 : 72
X Resolution                 : 72
Camera Model Name            : Redmi Note 14 Pro 5G
Software                     : MediaTek Camera Application
Image Description            :
Y Cb Cr Positioning          : Co-sited
Exif Version                 : 0220
Aperture Value               : 1.6
Mirror                       : false
Sensor Type                  : rear
Hdr                           : off
Op Mode                      : 36869
Small Picture                 : false
AI Scene                     : 0
Filter Id                    : 66048
```

```
CA\ Seleccionar Administrador: Símbolo del sistema

Filter Id                    : 66048
Zoom Multiple                : 1
Exposure Compensation        : 0
Exposure Program             : Program AE
Color Space                  : Uncalibrated
Max Aperture Value           : 1.6
Exif Image Height            : 4080
Brightness Value             : 0.7
Date/Time Original           : 2026:05:15 09:58:38
Flashpix Version             : 0100
Maker Note Byte Order        : Big-endian (Motorola, MM)
Sub Sec Time Original        : 203
White Balance                 : Auto
Interoperability Index       : R98 - DCF basic file (sRGB)
Interoperability Version     : 0100
Recommended Exposure Index   : 0
Exposure Mode                : Auto
Exposure Time                : 1/33
Offset Time                  : -04:00
Flash                        : Off, Did not fire
Sub Sec Time                 : 203
F Number                     : 1.6
Exif Image Width             : 1840
ISO                           : 800
Components Configuration     : Y, Cb, Cr, -
Offset Time Digitized        : -04:00
Focal Length In 35mm Format   : 23 mm
Sub Sec Time Digitized       : 203
Digital Zoom Ratio           : 0
Create Date                  : 2026:05:15 09:58:38
Shutter Speed Value          : 1/33
Metering Mode                 : Center-weighted average
Focal Length                 : 6.2 mm
Sensitivity Type              : ISO Speed
Offset Time Original         : -04:00
Scene Capture Type           : Standard
Light Source                  : Other
Resolution Unit               : inches
Xiaomi Model                  : Redmi Note 14 Pro 5G
Compression                  : JPEG (old-style)
Thumbnail Offset              : 1991
Thumbnail Length              : 38784
Profile CMM Type              :
```

Seleccionar Administrador: Símbolo del sistema

```
Profile CMM Type : 
Profile Version : 4.0.0
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
Profile Date Time : 2022:08:08 08:08:08
Profile File Signature : acsp
Primary Platform : Unknown ()
CMM Flags : Not Embedded, Independent
Device Manufacturer : Xiaomi
Device Model : 
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Apple Computer Inc.
Profile ID : 0
Profile Description : sRGB EOTF with DCI-P3 Color Gamut
Profile Copyright : Copyright Xiaomi Communications Co., Ltd. 2022
Media White Point : 0.96419 1 0.82489
Red Matrix Column : 0.51512 0.2412 -0.00105
Green Matrix Column : 0.29198 0.69225 0.04189
Blue Matrix Column : 0.1571 0.06657 0.78407
Chromatic Adaptation : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.01706
-0.00923 0.01508 0.75168
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width : 1840
Image Height : 4080
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 1.6
Image Size : 1840x4080
Megapixels : 7.5
Scale Factor To 35 mm Equivalent: 3.7
Shutter Speed : 1/33
Create Date : 2026:05:15 09:58:38.203-04:00
Date/Time Original : 2026:05:15 09:58:38.203-04:00
Modify Date : 2026:05:15 09:58:38.203-04:00
Thumbnail Image : (Binary data 38784 bytes, use -b option to extract)
t)
```

Tercer archivo de imagen

```
CAV Administrador: Símbolo del sistema
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool IMG_20260525_221040.jpg
ExifTool Version Number      : 13.59
File Name                    : IMG_20260525_221040.jpg.jpeg
Directory                   : .
File Size                    : 1354 kB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:06:17 15:14:59-04:00
File Access Date/Time        : 2026:06:17 15:36:22-04:00
File Creation Date/Time      : 2026:06:17 15:14:59-04:00
File Permissions              : -RW-RW-RW-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Make                         : Xiaomi
Orientation                  : Unknown (0)
Modify Date                   : 2026:05:25 22:10:40
GPS Latitude Ref              : North
GPS Speed                     : 0
GPS Altitude Ref              : Above Sea Level
GPS Processing Method         : network
GPS Speed Ref                 : km/h
GPS Version ID                : 2.2.0.0
GPS Longitude Ref             : West
GPS Time Stamp                : 02:10:31
GPS Date Stamp                : 2026:05:26
Y Resolution                  : 72
X Resolution                  : 72
Camera Model Name             : Redmi Note 14 Pro 5G
Software                     : MediaTek Camera Application
Image Description              :
Y Cb Cr Positioning           : Co-sited
Exif Version                  : 0220
Aperture Value                : 1.6
Mirror                        : false
Sensor Type                   : rear
Hdr                            : off
Op Mode                       : 32778
Small Picture                  : false
AI Scene                      : 0
Filter Id                     : 66048
Zoom Multiple                  : 1
```

```
CAV Seleccionar Administrador: Símbolo del sistema
Zoom Multiple                 : 1
Night Scene                   : 1
Exposure Compensation         : 0
Exposure Program              : Program AE
Color Space                   : Uncalibrated
Max Aperture Value            : 1.6
Exif Image Height             : 4080
Brightness Value              : 0.5
Date/Time Original            : 2026:05:25 22:10:40
Flashpix Version              : 0100
Maker Note Byte Order         : Big-endian (Motorola, MM)
Sub Sec Time Original         : 962
White Balance                  : Auto
Interoperability Index        : R98 - DCF basic file (sRGB)
Interoperability Version      : 0100
Recommended Exposure Index    : 0
Exposure Mode                  : Auto
Exposure Time                  : 1/24
Offset Time                    : -04:00
Flash                          : Off, Did not fire
Sub Sec Time                  : 962
F Number                       : 1.6
Exif Image Width              : 1840
ISO                            : 800
Components Configuration      : Y, Cb, Cr, -
Offset Time Digitized          : -04:00
Focal Length In 35mm Format    : 23 mm
Sub Sec Time Digitized         : 962
Digital Zoom Ratio             : 0
Create Date                    : 2026:05:25 22:10:40
Shutter Speed Value            : 1/24
Metering Mode                  : Center-weighted average
Focal Length                   : 6.2 mm
Sensitivity Type               : ISO Speed
Offset Time Original           : -04:00
Scene Capture Type             : Standard
Light Source                   : Other
Resolution Unit                : inches
Xiaomi Model                   : Redmi Note 14 Pro 5G
Compression                    : JPEG (old-style)
Thumbnail Offset               : 2021
Thumbnail Length               : 38784
Profile CMM Type               :
```

Seleccionar Administrador: Símbolo del sistema

```

Profile CMM Type           :
Profile Version            : 4.0.0
Profile Class              : Display Device Profile
Color Space Data           : RGB
Profile Connection Space   : XYZ
Profile Date Time          : 2022:08:08 08:08:08
Profile File Signature     : acsp
Primary Platform           : Unknown ()
CMM Flags                  : Not Embedded, Independent
Device Manufacturer        : Xiaomi
Device Model               :
Device Attributes          : Reflective, Glossy, Positive, Color
Rendering Intent           : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator            : Apple Computer Inc.
Profile ID                 : 0
Profile Description        : sRGB EOTF with DCI-P3 Color Gamut
Profile Copyright          : Copyright Xiaomi Communications Co., Ltd. 2022
Media White Point          : 0.96419 1 0.82489
Red Matrix Column          : 0.51512 0.2412 -0.00105
Green Matrix Column        : 0.29198 0.69225 0.04189
Blue Matrix Column         : 0.1571 0.06657 0.78407
Chromatic Adaptation      : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.01706
-0.00923 0.01508 0.75168
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width               : 1840
Image Height              : 4080
Encoding Process           : Baseline DCT, Huffman coding
Bits Per Sample            : 8
Color Components           : 3
Y Cb Cr Sub Sampling       : YCbCr4:2:0 (2 2)
Aperture                   : 1.6
Image Size                 : 1840x4080
Megapixels                 : 7.5
Scale Factor To 35 mm Equivalent: 3.7
Shutter Speed              : 1/24
Create Date                : 2026:05:25 22:10:40.962-04:00
Date/Time Original         : 2026:05:25 22:10:40.962-04:00
Modify Date                : 2026:05:25 22:10:40.962-04:00
Thumbnail Image            : (Binary data 38784 bytes, use -b option to extract)

```

```

GPS Altitude               : 13.7 m Above Sea Level
GPS Date/Time              : 2026:05:26 02:10:31Z
GPS Latitude               : 18 deg 29' 20.02" N
GPS Longitude              : 69 deg 54' 13.27" W
Circle Of Confusion        : 0.008 mm
Field Of View              : 76.1 deg
Focal Length 35mm Equiv    : 6.2 mm (35 mm equivalent: 23.0 mm)
GPS Position               : 18 deg 29' 20.02" N, 69 deg 54' 13.27" W
Hyperfocal Distance        : 2.87 m
Light Value                : 3.0
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>

```

Cuarto archivo de imagen

```
CA. Administrador: Símbolo del sistema
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool IMG_20260528_135833.jpg
ExifTool Version Number      : 13.59
File Name                    : IMG_20260528_135833.jpg.jpeg
Directory                   : .
File Size                    : 2.3 MB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:06:17 15:15:02-04:00
File Access Date/Time        : 2026:06:17 15:36:22-04:00
File Creation Date/Time      : 2026:06:17 15:15:02-04:00
File Permissions             : -rw-rw-rw-
File Type                   : JPEG
File Type Extension          : jpg
MIME Type                   : image/jpeg
Exif Byte Order              : Little-endian (Intel, II)
Make                        : Xiaomi
Orientation                 : Unknown (0)
Modify Date                 : 2026:05:28 13:58:33
GPS Latitude Ref            : North
GPS Speed                   : 0
GPS Altitude Ref            : Above Sea Level
GPS Processing Method        : network
GPS Speed Ref               : km/h
GPS Version ID              : 2.2.0.0
GPS Longitude Ref           : West
GPS Time Stamp              : 17:58:29
GPS Date Stamp              : 2026:05:28
Y Resolution                : 72
X Resolution                : 72
Camera Model Name           : Redmi Note 14 Pro 5G
Software                    : MediaTek Camera Application
Image Description            :
```

```
Image Description           :
Y Cb Cr Positioning        : Co-sited
Exif Version                : 0220
Aperture Value              : 1.6
Mirror                     : false
Sensor Type                : rear
Hdr                         : off
Op Mode                    : 36869
Small Picture               : false
AI Scene                   : 0
Filter Id                  : 66048
Zoom Multiple               : 1
Exposure Compensation      : 0
Exposure Program           : Program AE
Color Space                 : Uncalibrated
Max Aperture Value         : 1.6
Exif Image Height          : 4080
Brightness Value           : 3.4
Date/Time Original         : 2026:05:28 13:58:33
Flashpix Version           : 0100
Maker Note Byte Order      : Big-endian (Motorola, MM)
Sub Sec Time Original      : 439
White Balance              : Auto
Interoperability Index     : R98 - DCF basic file (sRGB)
Interoperability Version   : 0100
Recommended Exposure Index : 0
Exposure Mode              : Auto
Exposure Time              : 1/60
Offset Time                : -04:00
Flash                     : Auto, Did not fire
Sub Sec Time               : 439
F Number                   : 1.6
Exif Image Width           : 1840
ISO                        : 250
Components Configuration   : Y, Cb, Cr, -
Offset Time Digitized      : -04:00
Focal Length In 35mm Format : 23 mm
Sub Sec Time Digitized     : 439
```



```

Sub Sec Time Digitized      : 439
Digital Zoom Ratio          : 0
Create Date                 : 2026:05:28 13:58:33
Shutter Speed Value         : 1/60
Metering Mode               : Center-weighted average
Focal Length                : 6.2 mm
Sensitivity Type            : ISO Speed
Offset Time Original        : -04:00
Scene Capture Type          : Standard
Light Source                : Other
Resolution Unit             : inches
Xiaomi Model                : Redmi Note 14 Pro 5G
Compression                 : JPEG (old-style)
Thumbnail Offset            : 1991
Thumbnail Length            : 38784
Profile CMM Type            :
Profile Version             : 4.0.0
Profile Class               : Display Device Profile
Color Space Data            : RGB
Profile Connection Space    : XYZ
Profile Date Time           : 2022:08:08 08:08:08
Profile File Signature      : acsp
Primary Platform            : Unknown ()
CMM Flags                   : Not Embedded, Independent
Device Manufacturer         : Xiaomi
Device Model                :
Device Attributes           : Reflective, Glossy, Positive, Color
Rendering Intent            : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator             : Apple Computer Inc.
Profile ID                  : 0
Profile Description         : sRGB EOTF with DCI-P3 Color Gamut
Profile Copyright           : Copyright Xiaomi Communications Co., Ltd. 2022
Media White Point           : 0.96419 1 0.82489
Red Matrix Column           : 0.51512 0.2412 -0.00105
Green Matrix Column         : 0.29198 0.69225 0.04189
Blue Matrix Column          : 0.1571 0.06657 0.78407
Blue Matrix Column          : 0.1571 0.06657 0.78407
Chromatic Adaptation        : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.01706
                             : -0.00923 0.01508 0.75168
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width                 : 1840
Image Height                : 4080
Encoding Process            : Baseline DCT, Huffman coding
Bits Per Sample             : 8
Color Components            : 3
Y Cb Cr Sub Sampling        : YCbCr4:2:0 (2 2)
Aperture                    : 1.6
Image Size                  : 1840x4080
Megapixels                  : 7.5
Scale Factor To 35 mm Equivalent: 3.7
Shutter Speed               : 1/60
Create Date                 : 2026:05:28 13:58:33.439-04:00
Date/Time Original          : 2026:05:28 13:58:33.439-04:00
Modify Date                 : 2026:05:28 13:58:33.439-04:00
Thumbnail Image             : (Binary data 38784 bytes, use -b option to extract)
GPS Altitude                : 14.7 m Above Sea Level
GPS Date/Time               : 2026:05:28 17:58:29Z
GPS Latitude                : 18 deg 29' 13.65" N
GPS Longitude               : 69 deg 53' 50.74" W
Circle Of Confusion         : 0.008 mm
Field Of View               : 76.1 deg
Focal Length 35mm Equiv     : 6.2 mm (35 mm equivalent: 23.0 mm)
GPS Position                : 18 deg 29' 13.65" N, 69 deg 53' 50.74" W
Hyperfocal Distance         : 2.87 m
Light Value                 : 6.0

C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>_

```

Quinto archivo de imagen

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool IMG_20260615_084155.jpg
ExifTool Version Number      : 13.59
File Name                    : IMG_20260615_084155.jpg.jpeg
Directory                   : .
File Size                    : 2.7 MB
Zone Identifier              : Exists
File Modification Date/Time   : 2026:06:17 15:15:04-04:00
File Access Date/Time        : 2026:06:17 16:34:54-04:00
File Creation Date/Time      : 2026:06:17 15:15:03-04:00
File Permissions             : -rw-rw-rw-
File Type                   : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order              : Little-endian (Intel, II)
Make                        : Xiaomi
Orientation                  : Unknown (0)
Modify Date                  : 2026:06:15 08:41:55
GPS Latitude Ref             : North
GPS Speed                    : 0
GPS Altitude Ref             : Above Sea Level
GPS Processing Method         : network
GPS Speed Ref                : km/h
GPS Version ID               : 2.2.0.0
GPS Longitude Ref            : West
GPS Time Stamp               : 12:41:45
GPS Date Stamp               : 2026:06:15
Y Resolution                  : 72
X Resolution                  : 72
Camera Model Name            : Redmi Note 14 Pro 5G
Software                     : MediaTek Camera Application
Image Description            :
Y Cb Cr Positioning          : Co-sited
```

```
Image Description            :
Y Cb Cr Positioning         : Co-sited
Exif Version                 : 0220
Aperture Value               : 1.6
Mirror                       : false
Sensor Type                  : rear
Hdr                           : off
Op Mode                      : 36869
Small Picture                : false
AI Scene                     : 0
Filter Id                    : 66048
Zoom Multiple                : 1
Exposure Compensation        : 0
Exposure Program             : Program AE
Color Space                  : Uncalibrated
Max Aperture Value           : 1.6
Exif Image Height            : 4080
Brightness Value             : 7.3
Date/Time Original           : 2026:06:15 08:41:55
Flashpix Version             : 0100
Maker Note Byte Order        : Big-endian (Motorola, MM)
Sub Sec Time Original        : 995
White Balance                 : Auto
Interoperability Index       : R98 - DCF basic file (sRGB)
Interoperability Version     : 0100
Recommended Exposure Index   : 0
Exposure Mode                : Auto
Exposure Time                : 1/531
Offset Time                   : -04:00
Flash                        : Auto, Did not fire
Sub Sec Time                 : 995
F Number                     : 1.6
Exif Image Width             : 1840
ISO                          : 50
```

```
Administrador: Símbolo del sistema
ISO : 50
Components Configuration : Y, Cb, Cr, -
Offset Time Digitized : -04:00
Focal Length In 35mm Format : 23 mm
Sub Sec Time Digitized : 995
Digital Zoom Ratio : 0
Create Date : 2026:06:15 08:41:55
Shutter Speed Value : 1/531
Metering Mode : Center-weighted average
Focal Length : 6.2 mm
Sensitivity Type : ISO Speed
Offset Time Original : -04:00
Scene Capture Type : Standard
Light Source : Other
Resolution Unit : inches
Xiaomi Model : Redmi Note 14 Pro 5G
Compression : JPEG (old-style)
Thumbnail Offset : 1991
Thumbnail Length : 38784
Profile CMM Type :
Profile Version : 4.0.0
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
Profile Date Time : 2022:08:08 08:08:08
Profile File Signature : acsp
Primary Platform : Unknown ()
CMM Flags : Not Embedded, Independent
Device Manufacturer : Xiaomi
Device Model :
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Apple Computer Inc.
Profile ID : 0
Profile Description : sRGB EOTF with DCI-P3 Color Gamut
Profile Copyright : Copyright Xiaomi Communications Co., Ltd. 2022
Media White Point : 0.96419 1 0.82489
Red Matrix Column : 0.51512 0.2412 -0.00105
Green Matrix Column : 0.29198 0.69225 0.04189
Blue Matrix Column : 0.1571 0.06657 0.78407
Chromatic Adaptation : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.01706
```

```
-0.00923 0.01508 0.75168
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to extract)
Image Width : 1840
Image Height : 4080
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 1.6
Image Size : 1840x4080
Megapixels : 7.5
Scale Factor To 35 mm Equivalent: 3.7
Shutter Speed : 1/531
Create Date : 2026:06:15 08:41:55.995-04:00
Date/Time Original : 2026:06:15 08:41:55.995-04:00
Modify Date : 2026:06:15 08:41:55.995-04:00
Thumbnail Image : (Binary data 38784 bytes, use -b option to extract)
GPS Altitude : 13 m Above Sea Level
GPS Date/Time : 2026:06:15 12:41:45Z
GPS Latitude : 18 deg 29' 25.42" N
GPS Longitude : 69 deg 54' 3.16" W
Circle Of Confusion : 0.008 mm
Field Of View : 76.1 deg
Focal Length 35mm Equiv : 6.2 mm (35 mm equivalent: 23.0 mm)
GPS Position : 18 deg 29' 25.42" N, 69 deg 54' 3.16" W
Hyperfocal Distance : 2.87 m
Light Value : 11.5
```

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>_
```


Primer PDF analizado

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool "LISTA DE SERV CMCVV
 2 ENERO.pdf"
ExifTool Version Number      : 13.59
File Name                    : LISTA DE SERV CMCVV  2 ENERO.pdf
Directory                    : .
File Size                    : 249 kB
File Modification Date/Time   : 2026:01:06 15:20:05-04:00
File Access Date/Time        : 2026:06:17 15:36:21-04:00
File Creation Date/Time      : 2026:06:17 15:36:19-04:00
File Permissions              : -rw-rw-rw-
File Type                    : PDF
File Type Extension          : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.5
Linearized                   : No
Page Count                   : 1
Language                     : es-DO
Tagged PDF                   : Yes
Author                       : SALOMON Y ANA
Create Date                  : 2026:01:06 15:20:04-04:00
Modify Date                  : 2026:01:06 15:20:04-04:00
Producer                     : Microsoft« Excel« 2013
Creator                      : Microsoft« Excel« 2013
```

Segundo PDF analizado

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool "Levantamiento de ausencia de cámaras teleferico.pdf"
ExifTool Version Number      : 13.59
File Name                    : Levantamiento de ausencia de cámaras teleferico.pdf
Directory                   : .
Warning                      : FileName encoding must be specified [x2]
File Size                    : 1291 kB
File Modification Date/Time   : 2026:06:02 11:39:18-04:00
File Access Date/Time        : 2026:06:17 16:44:50-04:00
File Creation Date/Time       : 2026:06:17 16:44:49-04:00
File Permissions              : -rw-rw-rw-
File Type                    : PDF
File Type Extension           : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.4
Linearized                   : No
Page Count                   : 3
Language                     : es-DO
Tagged PDF                   : Yes
XMP Toolkit                   : 3.1-701
Producer                     : Microsoft® Word 2013
Creator                      : Cuenta Microsoft
Creator Tool                  : Microsoft® Word 2013
Create Date                   : 2026:06:02 11:39:17-04:00
Modify Date                   : 2026:06:02 11:39:17-04:00
Document ID                   : uuid:C28CDDF0-A0D7-4C9C-B01D-6D9D6171444E
Instance ID                   : uuid:C28CDDF0-A0D7-4C9C-B01D-6D9D6171444E
Part                          : 1
Conformance                   : A
Author                       : Cuenta Microsoft
```

Este es un comando para saber si está instalado y la versión del Exiftool, y también realice una búsqueda personalizada de los parámetros requeridos para cierta búsqueda.

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool -ver  
13.59
```

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool -CreateDate -Model -GPSPosition IMG_20260515_095837.jpg.jpeg  
Create Date           : 2026:05:15 09:58:38  
Camera Model Name     : Redmi Note 14 Pro 5G  
GPS Position          : 18 deg 29' 19.73" N, 69 deg 54' 13.14" W
```

Aquí aplicamos el borrado de metadatos de la segunda imagen y se puede observar que la mayoría de ellos ya no están, además de que tampoco los mas críticos aparecen

```
C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool -all= IMG_20260515_095837.jpg.jpeg
Warning: ICC_Profile deleted. Image colors may be affected - IMG_20260515_095837.jpg.jpeg
    1 image files updated

C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>exiftool IMG_20260515_095837.jpg.jpeg
ExifTool Version Number      : 13.59
File Name                    : IMG_20260515_095837.jpg.jpeg
Directory                   : .
File Size                   : 2.7 MB
File Modification Date/Time  : 2026:06:17 17:01:56-04:00
File Access Date/Time       : 2026:06:17 17:01:57-04:00
File Creation Date/Time     : 2026:06:17 15:15:04-04:00
File Permissions             : -rw-rw-rw-
File Type                   : JPEG
File Type Extension         : jpg
MIME Type                   : image/jpeg
Image Width                 : 1840
Image Height                : 4080
Encoding Process            : Baseline DCT, Huffman coding
Bits Per Sample             : 8
Color Components            : 3
Y Cb Cr Sub Sampling       : YCbCr4:2:0 (2 2)
Image Size                  : 1840x4080
Megapixels                  : 7.5

C:\Users\administrado\Desktop\LRRV\exiftool-13.59_64>
```

Estos son los metadatos principales que se obtienen con Exiftool

Geolocalización

- Latitud GPS
- Longitud GPS
- Altitud GPS
- Rumbo de la imagen GPS
(Dirección a la que apuntaba la cámara)
- Sello de tiempo GPS (Hora UTC sincronizada por satélite)

Fechas y Tiempos

- Fecha y hora de creación del contenido
- Fecha y hora de modificación del archivo
- Fecha y hora de digitalización
- Fecha y hora de acceso al archivo
- Zona horaria del dispositivo

Configuración Técnica de Captura

- Tiempo de exposición (Velocidad de obturación)
- Apertura del diafragma (Número F)
- Sensibilidad ISO
- Distancia focal
- Uso de flash (Activado/Desactivado)
- Modo de medición de luz
- Balance de blancos

Propiedades y Estructura del Archivo

- Tipo de archivo real
- Tipo MIME
- Tamaño del archivo
- Ancho de la imagen
- Alto de la imagen
- Resolución horizontal y vertical
- Compresión de datos
- Hash del archivo (identificadores únicos incorporados)

Identificación de Dispositivos y Hardware

- Marca de la cámara/móvil
- Modelo del dispositivo
- Número de serie del cuerpo de la cámara
- Número de serie de la lente
- Versión del firmware del dispositivo

Información de Vídeo

- Códec de vídeo
- Códec de audio
- Bitrate
- Frames por segundo (FPS)
- Duración
- Resolución
- Rotación
- GPS por fotograma
- Fecha de grabación

Información de Derechos y Autoría

- Autor
- Copyright
- Propietario
- Organización
- Información de contacto
- Licencia de uso
- Créditos IPTC

Información de Integridad y Forense

- Hash MD5
- Hash SHA-1
- Hash SHA-256
- Estructura de segmentos JPEG
- Tablas EXIF dañadas o alteradas
- Inconsistencias entre EXIF, XMP e IPTC
- Evidencias de edición o recompresión
- Firmas de software de manipulación

Con esta practica aprendí:

A utilizar la herramienta “**Exiftool**” para la obtención de los metadatos de una imagen o documento.

Ojo: si la imagen fue descargada de una red social lo mas probable es que no contenga la mayoría de los metadatos, como por ejemplo la imagen a continuación:

C:\Users\administrado\Desktop X

+

▼

```
ExifTool Version Number      : 13.59
File Name                    : WhatsApp Image 2026-06-14 at 7.15.36 PM.jpeg
Directory                    : C:/Users/administrado/Desktop/LRRV/exiftool-13.59_64
File Size                    : 191 kB
Zone Identifier               : Exists
File Modification Date/Time   : 2026:06:14 19:15:51-04:00
File Access Date/Time        : 2026:06:14 19:16:14-04:00
File Creation Date/Time      : 2026:06:14 19:16:06-04:00
File Permissions              : -rw-rw-rw-
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                     : image/jpeg
JFIF Version                  : 1.01
Resolution Unit               : None
X Resolution                  : 1
Y Resolution                  : 1
Image Width                   : 1600
Image Height                  : 721
Encoding Process               : Progressive DCT, Huffman coding
Bits Per Sample                : 8
Color Components               : 3
Y Cb Cr Sub Sampling          : YCbCr4:2:0 (2 2)
Image Size                    : 1600x721
Megapixels                    : 1.2
-- press ENTER --
```